Uncrossing Wires for 3-SAT: A Potential That Closes the Recursion

The Proof

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OpenAI and Claude (Anthropic) first announced this theorem. This note gives a short proof. The ideas behind it, the comparison with earlier work and examples are in the full paper. The Lean 4 library `UncrossingWires` in the same repository formalizes the argument down to a multitape machine, with other constants and another proof of the branching step.

Theorem 1. *For each $\varepsilon > 0$, 3-SAT with n variables is decided deterministically in time $C_\varepsilon |\varphi|^{O_\varepsilon(1)} 2^{\varepsilon n}$, where $|\varphi|$ is the size of the input φ.*

A statement in square brackets stands for 1 if it holds and 0 otherwise, and $[K] = \{0, \dots, K-1\}$. *Cauchy* is $(\sum_{i\in I} c_i d_i)^2 \le \sum_i c_i^2 \sum_i d_i^2$.

1 Networks

A *network* has *nodes* $s \in [K]$ and *weights* $w(s,t) = w(t,s) \ge 0$. A family B has numbers $B(s,t) < 2^{w(s,t)}$, and a *node function* F_s reads the numbers $B(s,\cdot)$; its values are dyadic rationals. The *value* is

$$\mathrm{val}(K, w, F) = \sum_B \prod_{s,t\in[K]} [B(s,t) = B(t,s)] \prod_{s\in[K]} F_s(B(s,\cdot)).$$

The bits of $B(s,t)$ are the *wires* between s and t. The potential is $\Phi(K,w) = K(\sum_{s,t} w(s,t)^2)^{1/2}$. By Cauchy over the K^2 ordered pairs,

$$\sum_{s,t\in[K]} w(s,t) \le \Phi(K,w). \tag{1}$$

2 One rectangle

Let x, y, u, v be bits, and $\hat{p} = (-1)^p$. Let $\chi_0(p,q) = \frac{1}{2}(1 + \hat{p}\hat{q})$, $\chi_1(p,q) = \frac{1}{2}(1 - \hat{p}\hat{q})$, $\chi_2(p,q) = \frac{1}{2}(\hat{p} + \hat{q})$, $\chi_3(p,q) = \frac{1}{2}(\hat{p} - \hat{q})$. So $\chi_0(p,q) = [p = q]$. Put $U = \chi_0(x,u)\chi_0(y,v)$ and $S = \chi_3(x,y)\chi_3(u,v)$. Then $[x = v][y = u] = U - S$, because both sides equal $\frac{1}{4}(1 + \hat{x}\hat{v} + \hat{y}\hat{u} + \hat{x}\hat{y}\hat{u}\hat{v})$. Moreover, by expanding in the $\hat{p}$,

$$2U = \sum_c \chi_c(x,y)\chi_c(u,v),$$
$$2S = \chi_0(x,u)\chi_0(y,v) - \chi_1(x,u)\chi_1(y,v) - \chi_2(x,u)\chi_2(y,v) + \chi_3(x,u)\chi_3(y,v).$$

Now let x, y, u, v be numbers of m bits, and U_j, S_j be U, S at the bit j. Then $[x = v][y = u] = \prod_j (U_j - S_j)$. Multiplying out, with T the set of positions that take $-S_j$, put

$$P^{\mathrm{r}} = \sum_{|T|\ge m/2} \prod_{j\in T} (-S_j) \prod_{j\notin T} U_j, \qquad P^{\mathrm{c}} = \sum_{|T|<m/2} \prod_{j\in T} (-S_j) \prod_{j\notin T} U_j,$$

so $P^{\mathrm{r}} + P^{\mathrm{c}} = [x = v][y = u]$. For $\pi \in \{0,1,2,3\}^m$ let $f_\pi(p,q) = \prod_j \chi_{\pi_j}(p_j, q_j)$.

Lemma 1. *Let $m = 0$, or $m = 2k+1$. Put $\beta_0 = 0$ and $\beta_{2k+1} = 2k + 1 + \lceil 8k/5 \rceil$. Each of $P^{\mathrm{r}}, P^{\mathrm{c}}$ is a sum over an index ξ of products $g_\xi(x,y)g'_\xi(u,v)$, and also a sum over an index ξ of products $g_\xi(x,u)g'_\xi(y,v)$. Across $x, y \mid u, v$, the index has β_m bits for P^{r} and $2m$ bits for P^{c}. Across $x, u \mid y, v$, it has $2m$ bits for P^{r} and β_m bits for P^{c}. All values are dyadic with $O(m+1)$ bits, computed in time polynomial in m.*

Proof. For $m = 0$: $P^{\mathrm{r}} = 1$ and $P^{\mathrm{c}} = 0$. Let $m = 2k+1$, and fix T. Across $x, y \mid u, v$, use S_j as it is, and the expansion of $2U_j$: the product of T is $(-1)^{|T|}2^{|T|-m}$ times the sum of $f_\pi(x,y)f_\pi(u,v)$ over the π with the

letter 3 on T. Across $x, u \mid y, v$, use U_j as it is, and the expansion of $2S_j$: the product of T is $2^{-|T|}$ times a signed sum of $f_\pi(x,u)f_\pi(y,v)$ over the π with the letter 0 off T. So each piece is $\sum_\pi c_\pi f_\pi f_\pi$ across either cut. A set T that fits π contributes a sign and a power of two that depend only on $|T|$ and π, and the fitting sets of one size are counted by one binomial, so c_π is a sum of at most $m+1$ binomials times signed powers of two. For $2m$ bits, name π with two bits per letter. For β_m bits: in P^{r} we have $|T| \geq k+1$, so across $x, y \mid u, v$ the string π has at most k letters other than 3; in P^{c} we have $|T| \leq k$, so across $x, u \mid y, v$ it has at most k letters other than 0. Then m bits mark the positions of the special letter, and $\lceil 8k/5 \rceil$ bits hold a number below 3^k with the other letters as base 3 digits; this fits because $3^{5k} = 243^k \leq 256^k = 2^{8k} \leq 2^{5\lceil 8k/5 \rceil}$. Other indexes get the zero function. □

3 One step

Let $(K'K', w, F)$ be a network, with nodes (a,b), $a, b \in [K']$. Rows $a < a'$ and columns $b < b'$ define a *rectangle* ρ, with *diagonals* $\{(a,b),(a',b')\}$ and $\{(a,b'),(a',b)\}$ of weights w_1, w_2. Let m_ρ be the largest odd number at most $\min(w_1, w_2)$, or 0 if the minimum is 0. The lowest m_ρ bits of the numbers on the diagonals are *paired*: with

$$x = B((a,b),(a',b')), \quad v = B((a',b'),(a,b)), \quad y = B((a,b'),(a',b)), \quad u = B((a',b),(a,b')),$$

modulo 2^{m_ρ}, their tests are the crossing $[x = v][y = u]$. All other bits are *carried*. A *choice* σ picks one piece $P_\rho \in \{P^{\mathrm{r}}, P^{\mathrm{c}}\}$ for each rectangle. There are at most $K'^4/4$ rectangles. For fixed B,

$$\prod_{s,t} [B(s,t) = B(t,s)] = \sum_\sigma \prod_{s,t} [\text{the carried bits of } B(s,t) \text{ and } B(t,s) \text{ agree}] \prod_\rho P_\rho. \tag{2}$$

Fix σ. Let d^{r}_ρ be β_{m_ρ} if $P_\rho = P^{\mathrm{r}}$ and $2m_\rho$ otherwise, and d^{c}_ρ be β_{m_ρ} if $P_\rho = P^{\mathrm{c}}$ and $2m_\rho$ otherwise. Build a network on the K' rows. Between rows $a < a'$, it has new wires: for all columns b, b', a copy of the carried bits between (a,b) and (a',b'), which the node a tests against $B((a,b),(a',b'))$ and the node a' against $B((a',b'),(a,b))$; and for all ρ with these rows, the d^{r}_ρ bits of the index ξ of ρ, with the factor $g_\xi(x,y)$ at a and $g'_\xi(u,v)$ at a'. So $w^{\mathrm{r}}(a,a) = 0$ and

$$w^{\mathrm{r}}(a,a') = \sum_b w((a,b),(a',b)) + \sum_{b<b'} f^{\mathrm{r}}_\rho, \qquad f^{\mathrm{r}}_\rho = w_1 + w_2 - 2m_\rho + d^{\mathrm{r}}_\rho. \tag{3}$$

The new node function sums over the old numbers on the row a:

$$F^{\mathrm{r}}_a(B') = \sum_B \mathsf{T}_a \prod_b F_{(a,b)}(B((a,b),\cdot)), \tag{4}$$

where T_a is the product of these tests and factors, for all other rows, and of the tests $B(s,t) = B(t,s)$ for two nodes s, t of the row a. In the same way we get a network $(K', w^{\mathrm{c}}, F^{\mathrm{c}})$ on the columns, with the factors $g_\xi(x,u)$ and $g'_\xi(y,v)$ across the columns, and $f^{\mathrm{c}}_\rho = w_1 + w_2 - 2m_\rho + d^{\mathrm{c}}_\rho$.

Lemma 2. *For each σ pick the network on the rows or on the columns. Then* $\mathrm{val}(K'K', w, F)$ *is the sum over σ of the values of the picked networks.*

Proof. Insert (4) in the value of the network on the rows and reorder the sums. For fixed old B, the new numbers split into independent parts. The tests of the new network leave one new number for each pair $a < a'$. A part that copies carried bits has two tests, and its sum is the test that those bits agree. The index of ρ gives $\sum_\xi g_\xi(x,y)g'_\xi(u,v) = P_\rho$. The tests inside a row are the tests of the carried bits for pairs in one row. So we get the summand of (2). The columns are the same. □

4 The potential

Lemma 3. *Let $\gamma = \sqrt{24/25}$. For every σ,* $\min(\Phi(K', w^{\mathrm{r}}), \Phi(K', w^{\mathrm{c}})) \leq \gamma\, \Phi(K'K', w)$.

Proof. Write each f^{r}_ρ in (3) as $\frac{1}{2}f^{\mathrm{r}}_\rho + \frac{1}{2}f^{\mathrm{r}}_\rho$. Then (3) has K'^2 terms, so by Cauchy $w^{\mathrm{r}}(a,a')^2 \le K'^2(\sum_{b<b'}\frac{1}{2}(f^{\mathrm{r}}_\rho)^2 + \sum_b w((a,b),(a',b))^2)$. Sum over the ordered pairs $a \ne a'$, multiply by K'^2, and do the same for the columns:

$$\Phi(K',w^{\mathrm{r}})^2 + \Phi(K',w^{\mathrm{c}})^2 \le K'^4\Big(\sum_\rho\big((f^{\mathrm{r}}_\rho)^2 + (f^{\mathrm{c}}_\rho)^2\big) + \sum_{s\ne t \text{ in one row or one column}} w(s,t)^2\Big).$$

Fix ρ, with $w_1 \ge w_2$ and $m = m_\rho$. One of $d^{\mathrm{r}}_\rho, d^{\mathrm{c}}_\rho$ is $2m$, which gives $f = w_1 + w_2$. The other is β_m, which gives $f \le w_1 + \frac{9}{10}w_2$: if $m = 0$ then $w_2 = 0$; if $m = 2k+1$ then $10(f - w_1) \le 10(w_2 - m) + 16k + 8$, and $w_2 \in \{2k{+}1, 2k{+}2\}$ gives $16k{+}8 \le 9w_2$ or $16k{+}18 \le 9w_2$. By Cauchy, $(f^{\mathrm{r}}_\rho)^2 + (f^{\mathrm{c}}_\rho)^2 \le (2{+}1{+}\frac{81}{100})(w_1^2 + w_2^2) \le \frac{96}{25}(w_1^2 + w_2^2)$, and $w_1^2 + w_2^2$ is half of the sum of $w(s,t)^2$ over the four ordered pairs of ρ. So the right side above is at most $\frac{48}{25}K'^4\sum_{s,t} w(s,t)^2 = \frac{48}{25}\Phi(K'K', w)^2$, and the smaller potential has a square at most half of this. □

5 The chain

Let $\ell \ge 0$ and $K_j = 2^{2^{\ell-j}}$, so $K_\ell = 2$ and $K_j = K_{j+1}K_{j+1}$.

Theorem 2. *Let (K_0, w, F) be a network with potential Φ. Then $\mathrm{val}(K_0, w, F)$ is computed in time $2^{K_0^2/3+4\gamma^\ell\Phi}$ times a constant depending on ℓ, times a polynomial in Φ and the time and the bit length of one evaluation of a node function.*

Proof. Put $N_0 = (K_0, w, F)$. Given N_j, pick a choice σ_j, and let N_{j+1} be the one of its two networks with the smaller potential, and the rows if equal. By Lemma 2, $\mathrm{val}(N_0)$ is the sum of $\mathrm{val}(N_\ell)$ over all chains $(\sigma_j)_j$. By Lemma 3, $\Phi(N_j) \le \gamma^j\Phi$. The network N_j has at most $K_j^2/4$ rectangles, and these numbers fall at least by a factor 4, so there are at most $2^{K_0^2/3}$ chains. For one chain, computing the weights takes polynomial time and fixes which networks are kept, since a comparison of potentials is a comparison of sums of squares. By (1) a family of N_ℓ has at most $\gamma^\ell\Phi$ bits.

A node function of N_{j+1} is (4), a sum over numbers with $\sum_b\sum_t w((a,b),t)$ bits, a sum of K_{j+1}^3 weights of N_j. By Cauchy its square is at most $K_{j+1}^3\sum_{s,t} w(s,t)^2 = \Phi(N_j)^2/K_{j+1}$, so there are at most $E_j = \gamma^j\Phi/\sqrt{K_{j+1}}$ bits. Each term has one evaluation of T and K_{j+1} evaluations of node functions of N_j. So one evaluation of a node function of N_ℓ takes at most $\prod_{j<\ell}(K_{j+1}2^{E_j})$ evaluations of node functions of N_0 and as many of T. For $r = \ell - j \ge 1$ we have $\sqrt{K_{j+1}} = 2^{2^{r-2}} \ge 2^{r-1}$, so $\sum_{j<\ell} E_j \le \gamma^\ell\Phi\sum_{r\ge1}\gamma^{-r}2^{1-r} = 2\gamma^\ell\Phi/(2\gamma-1) \le 3\gamma^\ell\Phi$. Multiplying: $2^{K_0^2/3}$ chains, $2^{\gamma^\ell\Phi}$ families, and $2^{3\gamma^\ell\Phi}$ evaluations. By Lemma 1, T takes polynomial time. The number of wires never grows, since $\lceil 8k/5\rceil \le 2k$ gives $\beta_m \le 2m$, so it is at most Φ. If the values at level j have b_j bits, then $b_{j+1} \le E_j + K_{j+1}b_j + O(\sum_\rho(m_\rho+1))$, so all values have polynomially many bits. □

6 Proof of the theorem

Fix $\varepsilon > 0$. By the sparsification lemma of Impagliazzo, Paturi and Zane (*J. Comput. System Sci.* 63 (2001), p. 513), there is $\Delta \ge 3$ such that a CNF on n variables with at most three literals per clause is satisfiable if and only if one of at most $2^{\varepsilon n/2}$ such CNFs is, each with at most Δ occurrences per variable, computed in time $2^{\varepsilon n/2}$ times a polynomial. Take ℓ with $16\Delta\gamma^\ell \le \varepsilon/2$, and $K = 2^{2^\ell}$. Take one formula Ψ. Reject it if it has an empty clause, and delete tautologies and repeated literals. An *occurrence* is a pair (i, C) of a variable i and a clause C containing it; there are at most Δn.

Placement. Number the variables $0, \dots, n{-}1$, and let $\omega(i) = i$ modulo K, so a node holds at most $(n{+}K)/K$ variables and $\Delta(n+K)/K$ occurrences. Going through the clauses, let $\mathrm{load}(s,t)$ count the occurrences (i,C) placed so far with $\omega(i) = s$, $\omega(C) = t$, and put C on the node t that minimizes $\sum_{i\in C}\mathrm{load}(\omega(i),t)$. Summed over all t this is at most $3\Delta(n+K)/K$, so its minimum is at most $3\Delta(n+K)/K^2$, and placing C adds at most $3 \le \Delta$. So all loads are at most $\kappa = \Delta(3n + 4K^2)/K^2$.

Network. Each occurrence $o = (i, C)$ is a wire between $\omega(i)$ and $\omega(C)$, with a bit θ_o in $B(\omega(i), \omega(C))$. The function F_s is the product of the tests that θ_o is the same for all occurrences of each variable i with $\omega(i) = s$, and that each clause C with $\omega(C) = s$ is true when each literal reads the bit of its occurrence. Then $\mathrm{val}(K, w, F)$ counts the satisfying assignments of the variables that occur, so it is positive if and only if Ψ has a solution. Since $w(s,t) \le \mathrm{load}(s,t) + \mathrm{load}(t,s) \le 2\kappa$ and $\sum_{s,t} w(s,t) \le 2\Delta n$, we get $\Phi(K,w)^2 \le K^2 \cdot 2\kappa \cdot 2\Delta n = 4\Delta^2 n(3n + 4K^2) \le (4\Delta(n + K^2))^2$. Theorem 2 gives time $2^{K^2/3 + 16\Delta\gamma^\ell(n+K^2)}$ times a constant and a polynomial, which is at most a constant times $2^{\varepsilon n/2}$ and a polynomial; here $n \le |\varphi|$, since the input declares its variables. Multiplying by the number of formulas proves the theorem. □